

# CREDIT CARD FRAUD DETECTION & Transaction Analytics

End-to-End Data Analytics Project Report

|                  |                                                        |
|------------------|--------------------------------------------------------|
| Technology Stack | Python   SQL   Power BI                                |
| Domain           | Banking / Financial Services / Fraud Analytics         |
| Dataset          | Credit Card Transactions — 284,807 records, 31 columns |
| Dashboard        | Executive Dashboard + Fraud Analytics                  |

Purpose: transform a highly imbalanced credit-card transaction dataset into actionable fraud insights through data preparation, exploratory analysis, feature analysis, and interactive Power BI reporting.

Prepared for project presentation / portfolio documentation

# 1. Executive Summary

A concise overview of the business problem, analytical approach, and key outcomes.

This project analyzes credit card transaction data to understand fraudulent transaction behavior, identify patterns that distinguish fraud from normal activity, and present the findings through an interactive Power BI dashboard. The project combines Python-based data analysis, SQL-oriented business analysis, and Power BI visualization into an end-to-end analytics workflow.

The dataset contains 284,807 transactions, of which 492 are classified as fraudulent. This represents approximately 0.17% of all transactions, creating a severe class-imbalance problem. The dashboard therefore focuses not only on transaction counts, but also on fraud rate, transaction value, time-based behavior, amount ranges, and the statistical association of PCA-transformed features with the fraud class.

The final dashboard is organized into two analytical pages: an Executive Dashboard for high-level monitoring and a Fraud Analytics page for deeper investigation. Together, they provide an executive-friendly view of fraud exposure and transaction patterns while establishing a foundation for subsequent machine-learning work.

| Metric                     | Observed value  | Interpretation                            |
|----------------------------|-----------------|-------------------------------------------|
| Total Transactions         | 284,807 (≈285K) | Overall transaction population            |
| Fraud Transactions         | 492             | Transactions classified as fraud          |
| Fraud Rate                 | 0.17%           | Fraud share of transaction count          |
| Total Transaction Amount   | \$25.16M        | Total amount shown by the dashboard       |
| Average Transaction Amount | 88.35           | Average transaction value                 |
| Fraud Transaction Amount   | 60.13K          | Amount associated with fraud transactions |
| Fraud Amount %             | 0.239%          | Fraud share of total transaction value    |

## 2. Business Problem & Objectives

Financial institutions need to detect fraudulent transactions quickly while minimizing false alarms. Because genuine transactions greatly outnumber fraudulent ones, a useful analytics solution must identify subtle patterns rather than rely on simple transaction counts.

### Primary business questions

- How large is the fraud population compared with normal transaction activity?
- What is the fraud rate and how does it vary across time?
- Do fraudulent transactions show different transaction-value behavior?
- Which transaction amount ranges show higher fraud rates?
- Which PCA-transformed variables show stronger statistical association with fraud?
- What insights can be converted into fraud-monitoring or risk-scoring recommendations?

### Project objectives

- Clean and validate the transaction dataset before analysis.
- Perform exploratory and statistical analysis using Python.
- Analyze fraud patterns by amount, time, and transaction class.
- Create interactive Power BI dashboards for executive and analytical audiences.
- Translate analytical findings into practical fraud-monitoring recommendations.
- Provide a foundation for future classification-model development.

### 3. Dataset, Data Preparation & Analytical Methodology

The dataset contains 31 columns: Time, V1–V28, Amount, and Class. The V1–V28 variables are PCA-transformed/anonymized numerical features. The Class variable is the target: 0 represents a normal transaction and 1 represents a fraudulent transaction.

#### Data-quality checks performed

- Dataset dimensions and column structure were inspected using Pandas.
- Data types and descriptive statistics were reviewed.
- Missing-value checks were performed.
- Duplicate records were identified; 1,081 duplicate rows were present in the analyzed dataset.
- Class distribution was examined to quantify the severe fraud imbalance.

#### Analytical workflow

- Python: data loading, cleaning, EDA, time/amount analysis, correlation analysis and feature exploration.
- SQL: structured business analysis and aggregation layer for transaction/fraud questions.
- Power BI: KPI design, interactive slicers, trend analysis, amount analysis and executive storytelling.

#### Important methodological note

The dashboard's Hour analysis is derived from the elapsed-seconds Time field. Because the dataset does not provide a real-world calendar timestamp, hourly patterns should be treated as exploratory patterns within the dataset rather than as verified clock-time behavior.

## 4. Power BI Page 1 — Executive Dashboard

High-level fraud monitoring and transaction overview.

The first dashboard page is designed for an executive or management audience. It emphasizes the size of the transaction population, fraud exposure, transaction value, time-based fraud rate, and the strongest PCA feature associations identified during exploratory analysis.

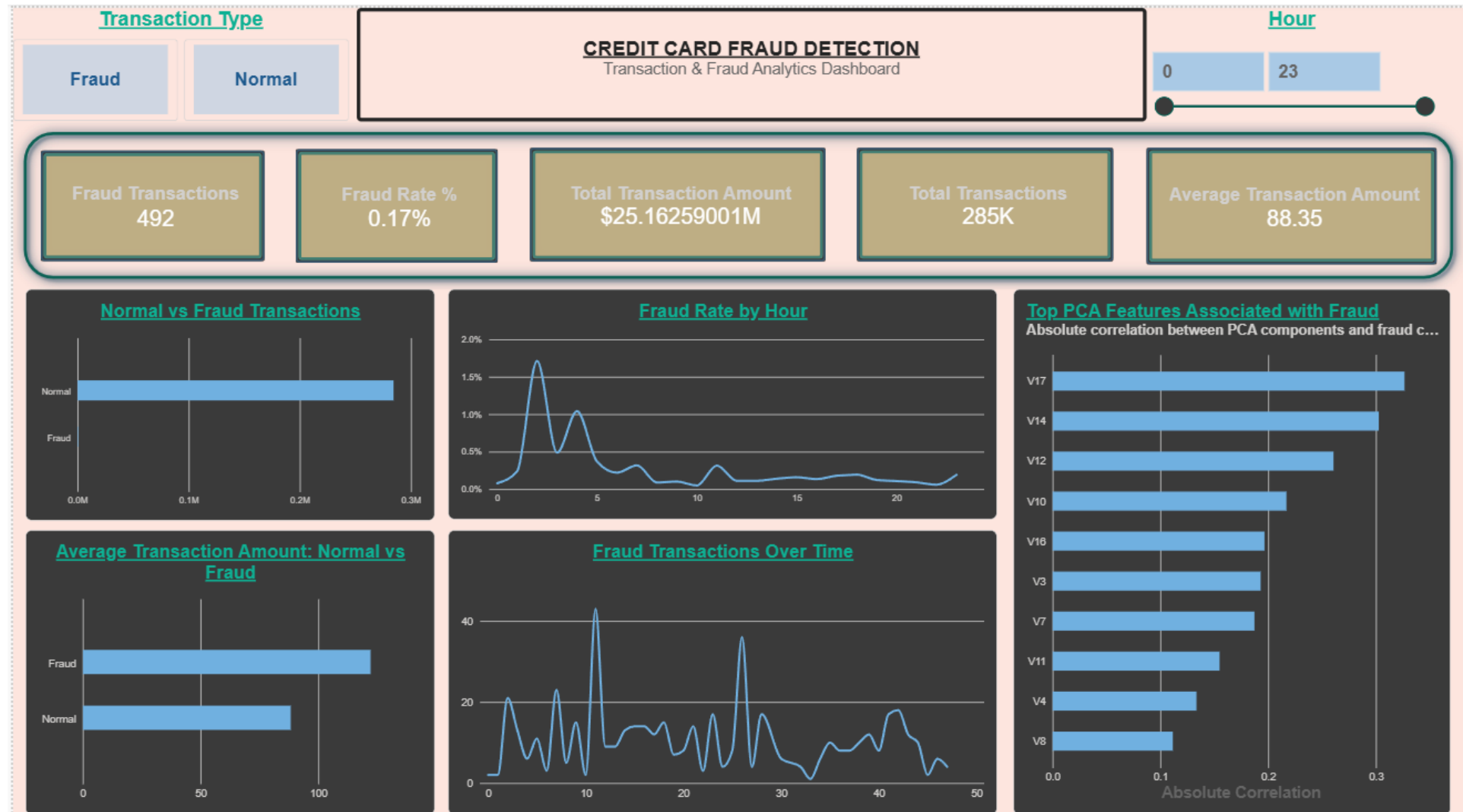


Figure 1. Executive Dashboard — Transaction & Fraud Analytics.

## 5. Executive Dashboard — Visual Interpretation

### KPI layer

- 492 fraud transactions are identified in the dataset.
- Fraud rate is approximately 0.17% of total transaction count.
- The dashboard reports approximately \$25.16M in total transaction amount.
- Average transaction amount is 88.35.

### Core visual insights

- Normal transactions dominate the transaction population, demonstrating the severe class imbalance.
- Fraud rate varies across hourly periods, with visible peaks in the line chart.
- Fraudulent transactions have a higher average transaction amount than normal transactions in this dataset.
- Fraud transactions fluctuate over the observation period rather than remaining evenly distributed.
- V17, V14, V12, V10 and other PCA components appear among the stronger features by absolute correlation with Class.

### Interpretation of PCA features

V1–V28 are PCA-transformed/anonymized variables. Their numerical relationship with Class can be analyzed, but they should not be given direct business labels such as income, location, merchant type, or customer risk. The PCA visual therefore communicates statistical association, not a causal or semantic business meaning.

# 6. Power BI Page 2 — Fraud Analytics

Detailed analysis of transaction patterns, risk indicators and fraud behavior.

The second page moves from executive monitoring to diagnostic analysis. It adds fraud amount exposure, fraud amount percentage, hourly transaction volume, fraud rate by transaction amount, fraud counts by hour, transaction distribution across amount ranges, and a key-findings section.

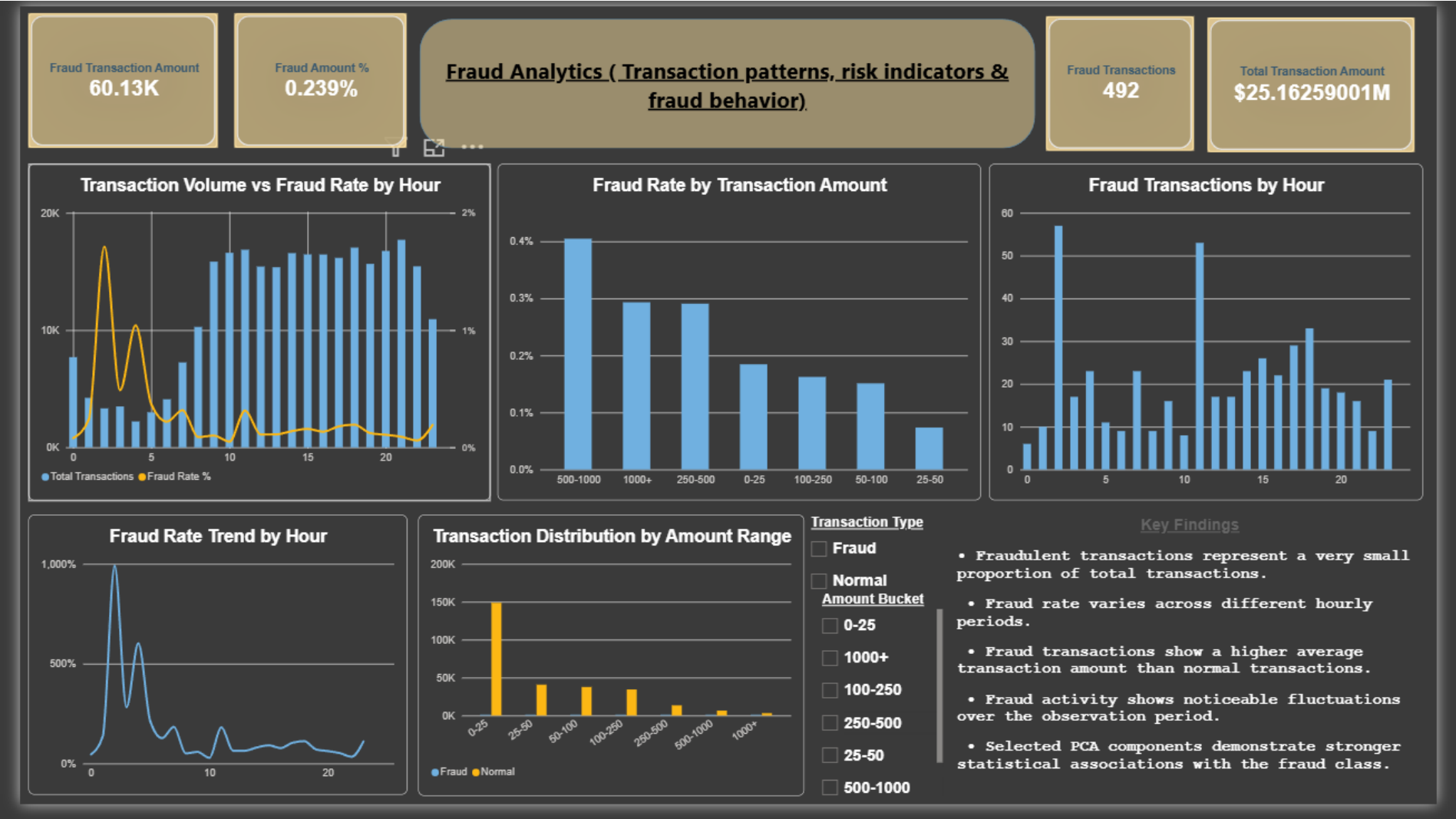


Figure 2. Fraud Analytics — Transaction patterns, risk indicators & fraud behavior.

## 7. Fraud Analytics Page — Detailed Findings

### Financial exposure

- Fraud Transaction Amount is shown as approximately 60.13K.
- Fraud Amount % is 0.239%, compared with a transaction-count fraud rate of 0.17%.
- The difference between transaction share and value share indicates that fraud exposure is not perfectly proportional to transaction volume.

### Amount-range analysis

- The dashboard shows variation in fraud rate across transaction amount buckets.
- The 500–1000 bucket shows the highest fraud rate among the displayed amount ranges, followed by 1000+ and 250–500.
- Lower amount ranges show comparatively lower fraud rates in the displayed analysis.

### Time-based analysis

- Transaction volume changes substantially across hourly periods.
- Fraud rate is not constant across hours and contains visible peaks.
- Fraud transaction counts also show spikes at specific hourly periods.

### Business interpretation

Amount and time are useful candidate risk signals, but neither should be used alone to automatically classify a transaction as fraudulent. A production fraud system should combine multiple behavioral and contextual signals and balance fraud recall against false positives.

## 8. Business Recommendations

### 1. Use multi-signal fraud scoring

Combine transaction amount, time behavior, historical transaction patterns, PCA features and other available contextual variables into a risk score instead of relying on a single threshold.

### 2. Monitor unusual fraud-rate spikes

Create automated monitoring for sudden increases in fraud count or fraud rate so investigators can prioritize potential fraud events.

### 3. Use amount as a supporting risk indicator

Higher-value transactions may deserve additional scrutiny, but amount should remain one signal within a broader decision framework.

### 4. Prioritize appropriate fraud metrics

Because fraud is rare, accuracy alone can be misleading. Precision, recall, F1-score, ROC-AUC and especially precision-recall analysis should be considered when a predictive model is developed.

### 5. Segment investigation workflows

Transactions can be prioritized using combinations of amount bucket, hour, fraud-risk score and other available customer/transaction context.

### 6. Establish dashboard-driven monitoring

Power BI can be used as a monitoring layer for management and analysts, while Python/ML services can provide predictive risk scores in a production architecture.

## 9. Technical Implementation

### Python analysis

- Pandas was used for loading, inspection and transformation of the CSV dataset.
- Class distribution, missing values, duplicates, amount behavior and time behavior were analyzed.
- Correlation analysis was used to rank PCA features by their association with Class.

### Power BI implementation

- DAX measures were created for Total Transactions, Fraud Transactions, Normal Transactions, Fraud Rate %, Total Transaction Amount and Average Transaction Amount.
- Additional DAX measures were created for Fraud Transaction Amount and Fraud Amount %.
- Calculated columns were used for Transaction Type, Hour, Amount Bucket and Amount Bucket Sort.
- Interactive slicers were added for Transaction Type, Hour and Amount Bucket.
- Two pages were designed: Executive Dashboard and Fraud Analytics.

### Core DAX measures

| Measure                    | Purpose                                                                  |
|----------------------------|--------------------------------------------------------------------------|
| Total Transactions         | Counts all transaction rows.                                             |
| Fraud Transactions         | Counts rows where Class = 1.                                             |
| Fraud Rate %               | $\text{Fraud Transactions} \div \text{Total Transactions}$ .             |
| Total Transaction Amount   | Sum of Amount across all transactions.                                   |
| Average Transaction Amount | Average of Amount.                                                       |
| Fraud Transaction Amount   | Sum of Amount where Class = 1.                                           |
| Fraud Amount %             | $\text{Fraud Transaction Amount} \div \text{Total Transaction Amount}$ . |

## 10. Limitations & Future Enhancements

### Current limitations

- The V1–V28 features are anonymized PCA components and therefore have limited direct business interpretability.
- Fraud transactions represent only about 0.17% of all transactions, making the problem highly imbalanced.
- The Time field is elapsed time rather than a full calendar timestamp.
- The dataset does not expose richer context such as merchant category, customer identity, device, location or account history.
- The current dashboard is an analytical reporting layer; predictive model performance is not represented on the current two pages.

### Recommended next phase

- Build baseline Logistic Regression and Random Forest fraud-classification models.
- Use stratified train/test splitting and appropriate class-imbalance techniques.
- Evaluate Precision, Recall, F1-score, ROC-AUC and Precision-Recall AUC.
- Tune the probability threshold according to the business cost of false positives vs missed fraud.
- Create a third Power BI page for ML Model Performance.
- Add model scores and risk bands to the dashboard if model outputs become available.

### Target future dashboard page

Page 3 — ML Model Performance: Confusion Matrix | Precision | Recall | F1-score | ROC-AUC | Precision-Recall Curve | Model Comparison | Risk Threshold Analysis.

# 11. Presentation / Viva Talking Points

## Problem statement

The objective is to analyze credit-card transactions and identify patterns associated with fraudulent activity so that financial institutions can improve fraud monitoring and prioritize suspicious transactions.

## Why is this project challenging?

Fraud is extremely rare compared with normal transactions. With only 0.17% fraud by transaction count, a model can achieve high accuracy while still missing fraud, so the analysis must focus on the right metrics and patterns.

## Why Power BI?

Power BI converts the analytical results into an interactive decision-support dashboard. Management can quickly see fraud exposure, trends, hourly patterns and amount-related risk signals.

## What does the PCA analysis mean?

V1–V28 are anonymized PCA-transformed features. The dashboard ranks them by absolute correlation with the fraud class to show which transformed dimensions have stronger statistical association with fraud.

## What is the most important business insight?

Fraud represents only 0.17% of transaction volume but approximately 0.239% of transaction value in the dashboard, and fraud rates vary by time and transaction amount.

## What would you do next?

The next step is to build and evaluate fraud-classification models using imbalance-aware methods and then bring model performance and risk scores into a third Power BI page.

## Conclusion

The project demonstrates an end-to-end data analytics approach to a realistic financial-services problem. The combination of Python analysis and Power BI reporting turns a highly imbalanced transaction dataset into a structured fraud-monitoring story. The current two-page dashboard provides a strong analytical foundation; adding model evaluation and risk scoring would complete the predictive analytics layer.

Portfolio positioning: Data Analyst | BI Analyst | Fraud Analytics | Financial Data Analytics